SCI-FLT-FIXED v0.1 Scientist Rationale

SCI-FLT-FIXED-SCIENTIST-RATIONALE v0.1/draft-r0.2

Status: implementation-blind Stage B explanatory draft; scientific-owner review required

Scientific owner: Grant Wilson

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Source SHA-256: 68b4a32c86fdbe6526af92c726ee5830c5300e7c9b5e2341f8bd43294abaf4eb
Shared normative core SHA-256: bbcce3546d895002af8f7bb847f32e734cb23e6b4963132a09e6f54ffe4d6481
Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
Owner directive SHA-256: 02ca63dcc8ac19e554fa333c20773ff85264336b4f622024605f6514ec1716d1
Build recipe SHA-256: 97b897beb2638d2b2081737fa8144a7e63d95f13eabde8f3bba774dc38a7cef5
```

Implementation-blind scientific-contract draft. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, performance, readiness, freeze, production, or Unity result.

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Normative import: the complete `SCI-FLT-FIXED-NORMATIVE-CORE v0.1/draft-r0.2`, source SHA-256 `bbcce3546d895002af8f7bb847f32e734cb23e6b4963132a09e6f54ffe4d6481`, is incorporated without modification. If this rationale and that core differ, the core controls.

1. The scientific question

SCI-FLT-FIXED answers one deliberately narrow question: given one exact admitted map-domain parent and one exact sampled convolution resolved before application under a fixed FLT-owned plan, what scientific product results when that convolution is applied once on the same grid?

The answer is the transformed parent-map amplitude

```
A_Theta,J = J_full L_Theta,
y = A_Theta,J m.
```

The restriction `J_full` matters as much as the convolution itself. It is resolved once before payload arithmetic from immutable parent membership and the exact required dependency footprint. The contract therefore describes a conditional linear transformation and the exact domain on which its result is scientific.

This is not a general filtering contract. It does not combine deterministic convolution with Wiener inference, matched or template-amplitude estimation, source learning, data-derived mode selection, automatic method selection, or per-realization relearning.

2. Why strict linearity is the base object

Strict linearity makes the estimand and its consequences explicit after parent membership and `J_full` are frozen. Doubling payload values on that domain doubles the transformed signal. An impulse exposes the exact sampled kernel. A compatible response propagates through the same `A_Theta,J`. An available covariance propagates by the same linear map on both sides.

This is not global linearity across parents with different support, validity, or membership. Recomputing a selector after a perturbation changes the applied object. Response perturbations, covariance draws, noise realizations, and NOI members therefore reuse the frozen selector; a missing required member value becomes unavailable rather than silently changing the row domain.

An additive offset would break this compact identity. It would need its own parent, units, support, uncertainty, reference-mode, response, covariance, NOI, lifecycle, and failure semantics. Rather than hide those scientific questions inside a familiar operation, v0.1 admits no additive term.

The word fixed applies to the entire scientific state, not merely to a numerical function call. Coefficients, parameters, support, normalization, grid, edge rule, transfer qualification, and the realized selector are frozen before the admitted payload or any companion realization is transformed.

3. Fixed convolution and the low-pass qualification

Convolution is the only numerically admitted base family in v0.1, not merely one example of an arbitrary dense linear family. `L_Theta` is its exact matrix representation. One FLT product applies that convolution once; a final kernel may be constructed elsewhere, but FLT makes no intermediate-composition claim.

The exact sampled coefficients, their grid offsets, center, phase, orientation, support, and normalization define the transformation. A family label or continuous ideal does not.

Low-pass is a further scientific claim about that exact sampled operator. It requires an exact coordinate-domain method, a compatible affine angular metric or another proof of angular transfer, frequency grid, DC gain, passband, transition, stopband or attenuation, phase, anisotropy, finite-grid and edge limitations, and complete parameter provenance. If any are absent, it can still be honest to say fixed convolution, but not low-pass.

The interior sampled-kernel transfer is also not automatically one global Fourier transfer for the finite row-restricted `A_Theta,J`. Edge restriction and missing-row selection break the symmetry required by that shortcut unless an exact theorem establishes otherwise.

This distinction prevents smoothing language from silently promising a frequency response that has never been stated. It also keeps a source-shaped kernel from being confused with a matched estimator: the former transforms a map; the latter estimates a template amplitude and belongs to a different scientific method.

4. Why the full-footprint rule is the sole v0.1 edge method

Near an edge or missing sample, several numerically convenient choices are possible. Extending the boundary, wrapping periodically, truncating the kernel, or renormalizing the surviving coefficients can all return finite numbers. They do not realize the same operator.

Truncation changes local gain and response. Support renormalization makes the operator position-dependent and changes noise and covariance. Reflection, inpainting, padding, and replacement add a boundary or missing-data model. Those can be scientifically defensible methods, but they require their own identities and contracts.

SCI-FLT-FIXED v0.1 therefore admits only rows with a complete authorized footprint. An edge row can remain in parent-shaped storage for alignment, but its scientific state is unavailable with a cause. It is not zero. This keeps numerical convenience separate from scientific admission.

The closure distinguishes geometric storage footprint `K_geom`, exact nonzero-coefficient support `K_nonzero`, and scientifically required dependency set `K_req`. Base v0.1 deliberately requires every stored geometric location, including an exact-zero coefficient; zero is a representation fact, not a floating threshold. This makes the admission rule reproducible and prevents a numerical tolerance from silently changing scientific support.

Identity and zero need explicit special cases. Identity requires only the same parent row and therefore preserves the exact admitted finite parent domain. The zero operator still inherits an exact declared parent-support row domain; an empty arithmetic support must not grant it every storage row. Numerical zero never erases parent identity, support, lifecycle, or unavailable companions.

5. What the transformed amplitude does and does not mean

The output is the transformed parent-map amplitude. Its units follow from the parent units and operator coefficient units. For a MAP parent, the originating nominal-beam identity and calibration lineage remain attached.

That does not make the result a new `mJy/filtered-beam` quantity. A unit-sum kernel preserves a constant field on complete support, but it does not by itself preserve a point-source peak, an aperture measurement, integrated flux, effective beam solid angle, or extended-source fidelity. Unit peak, unit angular integral, and unit L2 normalization answer different questions.

The contract therefore keeps stored transformed amplitude separate from response-corrected amplitude, fitted or template amplitude, uncertainty, standardized signal, and statistical significance.

6. Response, beam, transfer, and modes

Response must say what was perturbed. A fixed-state linear parent response, an already realized parent-grid response, and a parent full-procedure finite difference with FLT fixed can each receive the identical `A_Theta,J` exactly once, but they remain different response families. Re-resolving FLT during a procedure response leaves SCI-FLT-FIXED.

When no compatible basis, domain, or parent response exists, the requested transformed response remains unavailable. The kernel cannot stand in for the whole source response because the parent response, beam, centering, edge rule, and finite domain all matter. The exact zero operator has a local zero derivative, but that does not manufacture a complete sky-response identity or erase separately typed systematic uncertainty.

The local transfer and mode records describe the exact finite operator where those descriptions are scientifically defined. Nulling a local constant mode, for example, does not prove a whole-chain sky-mode claim. Influence similarly describes which parent rows and coefficients contribute; it is not physical exposure. An explicit exposure-lineage record carries the parent exposure identity or typed absence, states that FLT creates no physical exposure, and prevents a convolved exposure plane from being relabeled as the filtered signal's physical exposure.

7. Formal covariance and empirical uncertainty

If a compatible parent covariance is available, the exact deterministic propagation is

```
C_out = A_Theta,J C_parent transpose(A_Theta,J).
```

Even independent parent pixels generally become correlated after convolution because neighboring outputs reuse input pixels. A variance plane contains only marginals. It is not full covariance, and unknown cross terms are not zero.

The contract separates the parent stochastic authority from the output representation. A complete matrix propagated from an explicitly independent- diagonal parent model can be complete relative to that conditional model while remaining silent about unknown real parent cross terms. Every record therefore names its conditional model, representation, domain, rank, null space, omissions, supported operations, and excluded uncertainty classes.

Empirical uncertainty is a different ownership boundary. SCI-NOI applies the exact frozen FLT state to every compatible admitted randomization and infers the conditional uncertainty attachment. Filtering a variance, weight, standard deviation, standardized map, or significance field is not equivalent. Relearning a filter or selector for each member would define a different method and cannot be mixed into this fixed-state ensemble. A member that cannot supply a frozen required footprint becomes unavailable on that row.

8. Parent identity and ordering

A filtered MAP observation, a filtered MAP coadd, and a filtered JINC observation are different scientific products. Equal-looking arrays or WCS values do not erase their different estimands, membership, normalization, support, response, covariance, and lineage.

SCI-FLT-FIXED performs no coaddition and assumes no filter/coadd commutation. Such equality can hold only under explicit compatible operators, grids, weights, boundaries, normalization, support, and response facts. A future bounded proof could establish one relation without making it universal.

The admitted MAP and JINC boundaries also preserve upstream unavailability. At this launch state, the ordinary numerical MAP and TolTEC JINC routes remain gated upstream. Defining their successor types does not create numerical parents.

9. Lifecycle, atomicity, and honest absence

Requested, effective, resolved, applied, complete publication candidate, publication decision, and realized are not synonyms. The plan may disable the route. Required state may be unavailable. Application may fail. A transformed array may exist before its required response, support, validity, covariance-state, lineage, and failure records form an atomic bundle.

Publication policy evaluates the complete candidate, not a product that is already realized. Disabled yields `not_produced`. Identity and zero operators become real separately parented products only after the same candidate and publication sequence. A zero signal is not an unavailable row, a disabled route, zero total uncertainty, or infinite precision.

Honest absence is part of the product. A base transformed signal may carry an explicit unavailable response or covariance state where the role permits it. A response-qualified or covariance-qualified request cannot. Missing required records never become placeholders.

The SCI-NOI product is not an atomic FLT role. FLT carries only a typed NOI attachment-state relation. A separately parented NOI product may be attached later without deciding, reopening, or mutating FLT completeness.

10. Use and ownership limits

SCI-FLT-FIXED owns its local transformation, product, response state, support, validity, deterministic covariance state, lifecycle, and failure behavior. MAP or JINC continues to own the parent estimand. CAL owns absolute calibration. SCI-NOI owns empirical uncertainty inference. Beam, source, Pointing, OOF, and FRUIT interpretations remain with their respective future or upstream owners.

Three policy domains prevent bundle, row, and publication questions from being collapsed. Their draft records use VAL-congruent applicability, eligibility, realization, missing/conflict, lifecycle, and consumer-action states. They are not owner-approved Registry entries, and this draft claims no Registry evaluation.

The transformed product is therefore not a generic downstream admission ticket. A Beammap, Pointing, OOF, source-fit, catalog, NOI, or FRUIT consumer must provide an exact use policy. SCI-VAL may bind and evaluate an approved policy, but it does not invent producer facts or FLT policy.

11. Reading the falsifiable predictions

The core predictions turn the contract into observable distinctions:

- identity, zero, scaling, constant, impulse, signed, and zero-sum cases test the declared linear operator and normalization;
- footprint, missing-value, non-finite, and deferred-edge cases test the exact scientific row domain;
- response and covariance cases test composition without strengthening absent parent claims;
- WCS and parent-order cases test scientific identity rather than array similarity;
- NOI cases test exact fixed-state parity and rejection of relearning;
- lifecycle cases test disabled, unavailable, identity, zero, realized, and failed distinctions; and
- low-pass completeness and an independent sampled-transfer evaluation test whether a qualified transfer claim is actually supported;
- exact-zero coefficient and zero-operator support cases test required- dependency semantics independently of arithmetic nonzero support.

Passing these predictions would be evidence relevant to a later conformity or validation activity. This rationale reports no such result.

12. Resolution ownership and numerical evidence

"Externally resolved" means that one exact FLT-owned plan is complete before application. It does not mean an unnamed producer owns FLT policy or the transformation. A final kernel can arrive from elsewhere while FLT continues to own admission, the frozen selector, application, product identity, publication, and failure semantics.

Exact coefficients define scientific identity, but finite arithmetic needs a predeclared way to compare a candidate with an independent exact or high- precision oracle. The numerical-policy draft covers absolute, relative, and near-zero behavior, cancellation, conditioning, covariance, parallel agreement, overflow, underflow, non-finite values, and simultaneous row-level decisions. Freezing those rules before a result is seen prevents the tolerance from adapting to a failure. The policy is an evidence protocol, not new filter science or a validation result.

13. Stage B nonclaims

This rationale explains the draft scientific contract only. It makes no implementation-conformity, algorithm-change, validation, calibration, achieved-response, achieved-covariance, numerical-adequacy, performance, readiness, scientific-freeze, production, or Unity claim.